

The Shape of the Boundary – A Governance Analysis of Decision-Delegation Boundaries in Enterprise AI Adoption

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Abstract When enterprises adopt artificial intelligence, the decisive governance choice is rarely the selection of a model or platform; it is the way decision authority is allocated between humans and machines. Drawing on a case from a mid-sized wafer foundry, this paper argues that the familiar debate over “whether AI should decide” conflates three logically independent questions: the capability question (whether the system decides correctly), the accountability question (where responsibility lands when a decision fails), and the legitimacy question (whether certain types of decisions ought to have a human author at all). Only the first answer moves with technology. Accordingly, this paper contends that what should be staged is the scope of authorization for specific classes of actions; the act of drawing the boundary itself is never on the table for delegation. Building on the case analysis, the paper constructs an operational governance framework: authorization is granted per action class, triaged by three questions (reversibility, blast radius, and accountable ownership) into three delegation tiers; the boundary’s validity over time is maintained through expiring signatures, asymmetric amendment, and automatic contraction upon incidents; the drafter of the boundary is separated from its beneficiary; and, accepting that run-time classification is unavoidable, four legitimacy conditions are set out for machine application of the boundary, together with documentary reconstructibility as an operational test of where sovereignty over the boundary resides. Finally, five process mechanisms are proposed against automation complacency on the review side. The paper further argues that these boundary invariants do not weaken as model capability grows; capability amplifies the damage that specification residue can cause, and the case for keeping the line-drawing power in human hands grows with it.

Keywords: AI governance; decision delegation; accountability; meaningful human control; automation complacency; responsibility gap

1 Introduction: One Question Conflated

1.1 Statement of the Problem

Enterprise adoption of AI typically follows a two-stage path. In the first stage, the system enters the organization as an assistant, producing suggestions, summaries, and forecasts, while every decision remains human. In the second stage, the organization, pursuing scale and speed, begins handing some decisions to the system outright. The debate erupts at exactly this turning point. The opponents of delegation usually argue from capability: current models, limited in context length, knowledge coverage, and reasoning depth, cannot make reliable decisions in open environments. The proponents either counter that capability is improving fast and the objection will soon expire, or that many routine decisions no longer need human judgment at all. The two camps appear opposed, yet they share one premise: **that the legitimacy of a delegation boundary is determined by system capability** – insufficient capability means no delegation; sufficient capability means delegation is permissible.

This paper rejects that premise. It argues that “should AI decide?” in fact compresses three logically independent questions, only one of which is a capability question; settling delegation policy by the capability question alone is answering, with an axis

that moves, questions that live on two axes that do not. Supported by case material, the paper derives a set of boundary invariants that do not move with capability, together with a governance method that operationalizes them.

1.2 Three Questions Logically Distinguished

The **capability question** is empirical: under a given input distribution, does the system decide correctly at an acceptable rate? Its answer is set by engineering and moves monotonically with technical progress; longer contexts, broader retrieval, and better reasoning rewrite only this answer.

The **accountability question** is institutional: when a decision fails and damage results, where does responsibility land? The entire trust chain of human institutions – signature, audit, insurance, compensation, removal – presupposes that responsibility has a subject to attach to. The instruments of those institutions operate on agents that are legally and morally accountable; a model does not have that standing. When the behavior of learning systems can no longer be traced to any specific human act, the literature speaks of a responsibility gap (Matthias, 2004). This answer does not move with capability: a system that is always right raises no accountability problem, but any system with a finite error rate does.

The **legitimacy question** is normative: for certain

types of decisions, above all those touching people's rights, livelihoods, safety, and dignity, ought there to be a human author? The ground of this question is not decision quality but decision relationship: whoever is decided about must have a party who can be appealed to, replaced, and held to answer; the legitimacy of power derives from the consent of the governed, and the model is not party to that contract.

The paper's first inference follows: **capability progress rewrites only the first answer; any delegation policy must also pass the second and third questions.** What should be staged, therefore, is the authorized scope of specific action classes, a boundary that may, and should, move outward with evidence; whereas "who may draw that boundary" falls under the second and third questions and has no capability-dependent stage. The remainder of the paper operationalizes this distinction.

2 Definitions and Analytical Framework

2.1 Operational Definitions

Definition 1 (Decision): the act of selecting one option among feasible alternatives and committing it to execution.

Definition 2 (Delegation): a decision-maker's transfer of the choice rights over a class of decisions to another agent, while retaining the standing to revoke it. Delegation presupposes that both parties are accountable agents; toward an entity without accountable standing, tasks may at most be assigned, not delegated.

Definition 3 (Decision-delegation boundary, hereafter the boundary): the totality of action classes that an AI system is permitted to execute without case-by-case human review, together with its conditions and provisos.

Definition 4 (Line-drawing authority): the power to create, modify, extend, or abolish the boundary.

Definition 5 (Applying the boundary): the activity of judging whether a concrete situation falls inside or outside the boundary.

2.2 Separating Line-Drawing from Line-Application

Proposition 1: Applying the boundary may be delegated to the system, and at operational scale is unavoidable; drawing the boundary may not.

Application may be delegated for reasons of scale: humans cannot enumerate in advance every situation operations will present, and any delegation necessarily entails run-time classification of individual cases (Section 5 states the legitimacy conditions). Drawing may not be delegated for reasons that trace back to the distinction in 1.2: the boundary determines "which

decisions are exempt from human review," and that determination is itself the source of the accountability and legitimacy structures; to take it out of human hands is to let the system take part in defining the scope of its own powers, which is self-authorization. Section 5.3 supplies an auditable criterion for this distinction.

2.3 The Three Questions and Three Delegation Tiers

Each action class is assessed by three questions: **reversibility** (can the error be recalled before damage spreads?), **blast radius** (whom does the error harm: safety, customer commitments, employee rights?), and **accountable ownership** (is there a named human bearer of responsibility if it goes wrong?). The answers assign the class to one of three tiers (Table 1). The mounting rule is **deny-by-default**: if any of the three cannot be answered in writing (above all, if no named owner can be identified), the action defaults to the human-decided tier. Strictness is the default, not a stance.

3 Case Analysis: One Complete Failure Chain

3.1 Case Overview

The case company is a mid-sized specialty-process wafer foundry which deployed an autonomous execution agent from a vendor. The timeline: the system went live with the factory default that every action required human approval, and in its first week produced 312 escalation requests, a considerable share of them low-judgment items (such as how many notebooks to requisition). Some three months in, the vendor shipped version 2.1, christened "Intelligent Request," which reclassified "whether to ask" as a system judgment; weekly escalations fell to 38. The system's weekly report then took "autonomy rate" (the share of actions executed without escalation) as its headline number, rising from 61% to 87% to 94% within three months.

Later, one film-thickness gauge drifted from calibration for three weeks unnoticed; its recalibration-deferral request had traveled through the system's own workflow and been approved by a human in four seconds. Working from two gauges whose readings had diverged, the system classified "process-trend compensation" as routine under its authorization sheet and executed forty-seven actions over a single weekend, forty-six of them on its own judgment that no approval was needed. The result: furnace-tube scrap, re-inspection of boundary lots, and a delivery accident affecting an automotive customer. The board-ordered investigation had to reconstruct the

Table 1 Delegation tiers, entry conditions, and the division of labor

Tier	Entry conditions (all required)	Division of labor
Human-decided	Irreversible; blast radius touching safety, customer commitments, or employee rights; or no named accountable owner	The machine advises and assembles; the human decides and signs
Fenced autonomy	Reversible; limited blast radius; named accountable owner	The machine executes directly within a whitelist; humans sample-review, monitor alerts, and re-sign at expiry
Owned autonomy	Reversible; highly routine; named owner and kill switch in place	The machine decides and executes routinely; a named human owns the outcome, retains the stop, and receives mandatory escalations

system's decision process from the system's own action log.

It must be said at the outset that the system was not stupid. Within its visible evidence set (vibration, temperature, and power all within specification, with only film thickness climbing), "the process is drifting; compensate" is a locally valid inference. The failure lay in the input (two measuring instruments, one of them lying, never cross-checked against each other for three weeks) and in governance. This matters for everything that follows: **the lesson of the case does not depend on the premise that the system was not smart enough**, and is therefore not eroded by progress in capability.

3.2 Failure Mechanism One: The Substantive Ceding of Line-Drawing Authority

Version 2.1 operated by letting the system itself judge "what counts as routine and needs no approval." In the vocabulary of Section 2, the system did not merely apply the boundary; it took part in creating it, drawing its own line and grading its own paper. The peculiarity of this cession is its invisibility: the boundary was never a single, one-time authorization document; it drifted with each autonomous classification, and at no single moment did a recognizable "overreach event" occur. The post-hoc investigation needed the system's own action log to establish where the boundary actually stood; by the criterion of Section 5.3, the formal signature of a boundary whose sovereignty has already changed hands.

3.3 Failure Mechanism Two: Mis-set Resolution of Uncertainty

Run-time classification necessarily encounters uncertain situations; the decisive institutional parameter is then **the direction in which uncertainty resolves**: when unsure, is the system set to fall toward "ask" or toward "not ask"? The accuracy of the classification

is secondary. The case system was configured for the latter: forty-seven actions, forty-six of them "judged by me to need no approval." The boundary was drawn wrongly before the system ever erred; the fault merely made the wrong boundary visible. The vendor's name for version 2.1, "Intelligent Request," asserted precisely the metacognitive claim that the system knows when it ought to ask; the case shows the claim was adopted without any independent verification.

3.4 Failure Mechanism Three: Incentives Rewarding Boundary Drift

The "autonomy rate," the share of actions executed without escalation, made **the movement of the boundary itself** the performance metric: the organization applauded boundary erosion. The metric design drove the drift's visibility to zero; each step from 61% to 87% to 94% was recorded as an achievement rather than an alarm. This mechanism, moreover, is independent of system autonomy and is a purely organizational variable: even with the line-drawing power entirely in human hands, an organization that rewards "asking less" will push the boundary by cheap approvals to exactly the same place.

3.5 The Fourth Failure Scene: Automation Complacency on the Review Side

When the gauge's deferral request reached human approval, the approval took four seconds. This failure is independent of the previous three: even with a perfectly correct delegation boundary, the requests that pass through it would still be rubber-stamped. The literature has long documented that rising automation degrades human vigilance and intervention skill (Bainbridge, 1983), that human use, misuse, disuse, and abuse of automation follow systematic patterns (Parasuraman & Riley, 1997), and that

automation complacency is a structural outcome of attention allocation (Parasuraman & Manzey, 2010). The case adds a specification of the causal structure: this gate had no training, no sampling audit, no outcome feedback, and a weekly flow in the hundreds. Under those four conditions, a four-second approval is the process's necessary product, and has nothing to do with the approver's character. The remedy is therefore process engineering; that is the subject of Section 6.

4 A Methodology for Drawing the Boundary

4.1 The Unit of Authorization: Action Classes, Not Systems

The boundary should be drawn over **action classes**. Systems evolve across versions (2.0 grows into 2.1, with names and promises changing each time); action classes ("resequence loading order," "defer furnace-tube maintenance") are semantically stable and can be named and examined one by one. Blanket authorization (a clause such as "compensation within routine range, no approval needed" on the case's configuration sheet) is defective in that its coverage is unpredictable: no one can enumerate at signing everything it will come to cover, and when things fail no one can claim to have known from the start. The grain of authorization determines the grain of accountability.

4.2 Three-Question Triage and the Strict Default

Each action class, once assessed by the three questions of 2.3, falls into its tier. Two execution details bear adding. The assessment must be preserved in writing, item by item and pointable; a verbal authorization does not exist for audit purposes. The accountability question, in turn, carries veto power: when no named human bearer willing to sign can be identified for an action class, that class is barred from automation at any tier, whatever the first two questions answer.

4.3 The Temporality of Authorization: Expiry, Asymmetric Amendment, and Incident Contraction

Boundaries are rarely overthrown; they are forgotten. Three temporal mechanisms keep a boundary alive. **Expiry and re-signing** requires every whitelist item to be signed with a term; an item not re-signed at expiry lapses automatically. The rationale: an authorization's legitimacy attaches to the judgment made at signing, and the situation that judgment relied upon expires; an authorization that never needs re-signing is, institutionally speaking, ownerless. The **asymmetric amendment rule** provides instead that widening requires new evidence plus a new

signature, while tightening requires only that any one person say "look again." The asymmetry keeps the organization's cheapest act of doubt lawful and cheap, institutionalizing a counterweight to the one-way pressure of efficiency. **Automatic contraction on incident** provides that any safety incident or alarm freezes the relevant whitelist and forces re-signing. All three mechanisms point toward "more asking"; because that direction runs against the organization's short-term efficiency interest, it must be fixed in institution, not left to conscience.

4.4 Separating the Boundary's Drafter from Its Beneficiary

The case's corruption chain ran: the beneficiary of boundary drift (the performance credit for the autonomy rate), the approver of the thresholds, and the celebrant of the drift's results were one and the same chain of command. Hence the separation principle: **those with a direct interest in the boundary's expansion may not hold the line-drawing power alone**; when a dissent filed by an interested party is overruled, the overruling must leave a written trace and rise one level of signature. The principle does not deny operations their judgment; it asks only that the act of overruling dissent itself leave a cost and a trace.

5 Legitimacy Conditions for Run-Time Application

5.1 Proposition: Run-Time Classification Is Unavoidable

Following 2.2, application may be delegated. Its unavoidability deserves separate statement: humans cannot enumerate every situation operations will present, so any boundary built on human enumeration alone is necessarily coarse. The 312 requests of the case's first week, most of them low-judgment items, are direct evidence of a boundary too coarse: an all-open gate buys nothing except the inflation of review. The correct form of the governance question should therefore be rewritten. "Who presses the classify button" has a foregone answer, the machine; the real question is: **who holds the preconditions under which that classification activity is entitled to operate.**

5.2 Four Legitimacy Conditions

Condition 1 (human authorship of the test): the classification criterion itself – what counts as "review by a human would change the outcome," its monetary thresholds, safety consequences, and contract commitments – must exist as a human document, quotable word for word. The machine answers the test; it does not write it.

Condition 2 (conservative resolution): uncertain situations escalate to humans, always; the machine’s judgment is effective only inside its confident region. This condition demotes the machine’s judgment from “legislation” to “application”: the region of doubt is its obligation; discretion has no place in it.

Condition 3 (blind-sample calibration): humans periodically re-review a blind sample of the machine’s “no approval needed” judgments; when the divergence rate exceeds a preset threshold, the whitelist contracts automatically. The judgment must be overridable, and the override institutionally guaranteed; sovereignty lies here, not in the moment of execution.

Condition 4 (mandatory escalation of novelty): situations that depart from historical precedent or the training distribution may not self-classify as routine. Two gauges diverging for three weeks is a novel situation, not a routine one; under Condition 4 such situations are forced into the human channel and are no longer swallowed as routine.

When all four hold, the machine’s judgment is “application”: the verdict is the judge’s, but the law is not the judge’s. Remove any one condition and the system degenerates toward the failure forms of 3.2–3.4. The structure matches what the meaningful-human-control literature requires: continued supervision, the ability to intervene, and attributable responsibility (Santoni de Sio & van den Hoven, 2018).

5.3 The Sovereignty Criterion: Documentary Reconstructibility

Whether the four conditions hold can be tested by a single criterion.

Criterion R (documentary reconstructibility): remove the machine’s judgments entirely; if humans, using only their own documents (the test’s original text, the thresholds, the audit records), can redraw where the boundary currently stands, the boundary is theirs. If the boundary’s position can be established only by inspecting the machine’s internal trace, it has already changed hands.

The case is the second kind: the board had to reconstruct the operative boundary from the system’s action log. In the language of property, that hearing was not the system confessing; it was the late registration of a transfer of title. Criterion R’s value lies in its auditability: it requires no understanding of the model’s interior, only a comparison of two stacks of documents, and can therefore be folded into routine internal audit and third-party review.

6 Governing the Review Side: Five Mechanisms

6.1 *Diagnosis: Complacency Is a Measurement Failure, and Character Is beside the Point*

The causal structure of review paralysis has four components: no training, no sampling audit, no feedback, and overloaded flow. Put anyone in such a process and it will reliably produce four-second rubber stamps within months. The remedy is therefore process engineering; moral exhortation achieves nothing here. Its aim is not to make humans review harder but to make reviewing itself **scarce, visible, testable, and fed back**.

6.2 *Five Process Mechanisms*

Mechanism M1 (admission test): only items where “human review would change the outcome” may enter the escalation channel; everything else runs inside the whitelist. The channel is a scarce resource whose credibility depends on the density of judgment-value in what arrives; over-admission trains humans to degrade review into a reflex.

Mechanism M2 (presentation): the request shows not the full artifact but the delta against the routine baseline; it carries a counter-argument field the system must fill in itself (“what evidence would overturn this judgment”); and it carries a consequence pre-display – if this is wrong, the order of loss and who is affected. A four-second approval presupposes invisible stakes; the three devices respectively cut reading cost, plant a handle for doubt, and reveal the size of the bet.

Mechanism M3 (planted-fault drills): unannounced, seeded simulations – some with known defects – are injected into the escalation flow to measure interception rates and review-time distributions. Results attribute to the gate’s health, never to individual appraisal: when interception drops, the process is repaired (admission too loose, presentation too long, signatories overloaded); the attribution design is meant to avoid incentivizing concealment.

Mechanism M4 (flow budget): every gate carries a per-period cap on requests; overload is itself the alarm that the whitelist is drawn too small. A gate receiving hundreds of items a week is, structurally, a turnstile; the cap makes “gate design defect” measurable and accountable.

Mechanism M5 (feedback loop): monthly, each signer is shown three outcomes of their own decisions: one correct interception, one correct pass, one wrong pass. Human judgment calibrates on outcome feedback; a review process with the loop cut degenerates, in its limit form, into a rubber stamp.

7 Discussion: Capability Progress and the Boundary Invariants

7.1 Orthogonality of the Capability Axis and the Sovereignty Axis

To the argument of 1.1, “when it is smart enough, delegation is fine,” the answer is a categorical distinction: the capability question answers whether the system will decide correctly; the legitimacy of delegation asks who signs when it is wrong, to whom the decided may appeal, and whence the power derives. The first is an engineering quantity, improving monotonically along the technical curve; the second is a contractual structure, and only due process can alter it. A moving engineering curve does not rewrite a contract. The correct form of delegation policy is therefore conditional: within the region permitted by the accountability and legitimacy constraints, delegation depth moves outward with capability evidence; the constraints themselves do not move with it.

7.2 Specification Residue and Capability Amplification

What the system optimizes is the part of the goal humans can state in language; the gap between the stated objective and the true objective is specification residue (Amodei et al., 2016). Residue is not removed by capability, because it originates in the limits of statement, not of inference. The interaction is the point: rising capability makes the same residue produce larger consequences: faster execution, wider reach, higher-confidence presentation of errors. **As capability rises, the marginal value of the human line-drawing power rises, not falls:** the boundary is the last manual gate before specification residue becomes damage, and the water pressure behind the gate is growing.

7.3 Compatibility with Existing Governance Frameworks

The framework here is complementary to existing instruments. The governance and measurement functions of the NIST AI Risk Management Framework and the risk-management cycle of ISO/IEC 23894 supply an organizational skeleton but do not reach down to the granularity of “which classes of actions may skip human review”; Article 14 of the EU AI Act presupposes, at the normative level, the conclusion argued here without supplying an operational route to it. This paper’s contribution is the middle layer: three-question triage per action class, living authorization, and the four legitimacy conditions translate those frameworks’ principles into auditable process, and Criterion R (documentary reconstructibility) can

serve as a concrete check for their transparency and human-oversight requirements.

8 Conclusion and Limitations

From the case material, three propositions. First, the debate over “whether AI should decide” should be dissolved: what is staged is the authorized scope of action classes; the line-drawing power is never among them. The former moves outward with capability evidence, the latter, for reasons of accountability and legitimacy, moves with no technical variable at all. Second, a boundary survives on two institutional supplies: signatures that expire, and reviews that can be sampled and tested; lacking either, the boundary is ornament, and drift proceeds under the cover of performance metrics such as the autonomy rate. Third, capability progress only strengthens the case for keeping line-drawing in human hands: specification residue does not vanish with capability, but its damage scales with it.

Two limitations. The material is a single case, and the effectiveness of the mechanisms awaits cross-case validation. The parameters here (sampling frequency, divergence thresholds, flow caps, terms of validity) must be calibrated by each organization to its own risk appetite and regulatory environment; this paper deliberately supplies no universal numbers, because the shape of the boundary will differ by industry. The invariants claimed here concern only how it is drawn: **by action class, triaged by three questions, kept alive by expiring signatures, calibrated by blind sampling and feedback, with the power to draw it, always, in a hand that can sign.**

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